

# On Modal Logic of Full Products of Neighborhood Frames

Rajab Aghamov<sup>1</sup>   Andrey Kudinov<sup>2,3</sup>   Maik Nguyen<sup>1</sup>   Jakob Piribauer<sup>1</sup>

<sup>1</sup>Technische Universität Dresden

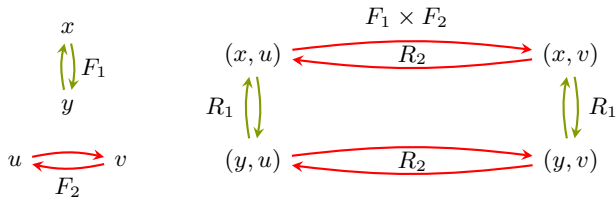
<sup>2</sup>Higher School of Modern Mathematics

<sup>3</sup>HSE University

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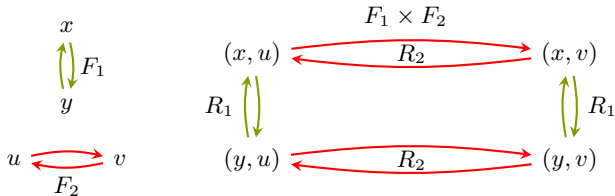
# Motivation

- Product of Kripke frames were studied by Shehtman and Segerberg in 70s



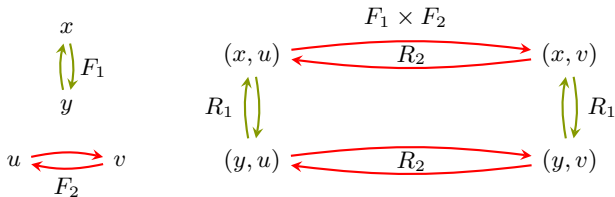
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- $\Box_1\Box_2p \leftrightarrow \Box_1\Box_2p$  (com) and  $\Diamond_1\Box_2p \rightarrow \Box_2\Diamond_1p$  (chr) are valid



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- $\Box_1 \Box_2 p \leftrightarrow \Box_1 \Box_2 p$  (com) and  $\Diamond_1 \Box_2 p \rightarrow \Box_2 \Diamond_1 p$  (chr) are valid
- $S4 \times S4 = S4 \otimes S4 + (\text{com}) + (\text{chr})$



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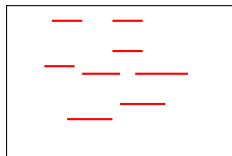


Figure: Element of horizontal topology

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- (com) and (chr) fails on products of topological frames
- $S4 \times_t S4 = S4 \otimes S4$
- $S4 \times_t^+ S4 = S4 \otimes S4 \otimes S4 + \Box p \rightarrow \Box_1 p \wedge \Box_2 p$

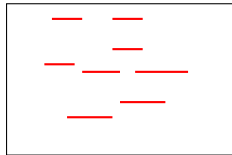


Figure: Element of horizontal topology



# Contribution

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- $S4 \times_t S4 = S4 \otimes S4$
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## Theorem

*Let  $\Lambda$  be a unimodal logic with  $\Lambda \in \{T, D\}$ . Then the following holds:*

$$\Lambda \times_n^+ \Lambda = \Lambda \otimes \Lambda \otimes \Lambda + \Box p \rightarrow \Box_1 \Box_2 p \wedge \Box_2 \Box_1 p$$

# Neighbourhood frame

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## Definition

Let  $X$  be a non-empty set and  $k \geq 1$ . A  $k$ -neighborhood frame (n-frame) is a pair  $(X, \tau)$ , where  $\tau : X \rightarrow 2^{2^X}$  assigns to every  $x \in X$  a *filter*  $\tau(x)$  on  $X$ , i.e.

- $\tau(x) \neq \emptyset$
- if  $U \in \tau(x)$  and  $U \subseteq V$  then  $V \in \tau(x)$
- if  $U, V \in \tau(x)$  then  $U \cap V \in \tau(x)$

# Full Product of Frames

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## Definition

Let  $\mathcal{X} = (X, \tau_1)$  and  $\mathcal{Y} = (Y, \tau_2)$  be two unimodal neighborhood frames. Then the full product of these two frames is a 3-n-frame  $\mathcal{X} \times_n \mathcal{Y} = (X \times Y, \tau'_1, \tau'_2, \tau)$  defined as follows :

$$\tau'_1(x, y) = \{U \subseteq X \times Y \mid \exists V \in \tau_1(x) : V \times \{y\} \subseteq U\},$$

$$\tau'_2(x, y) = \{U \subseteq X \times Y \mid \exists V \in \tau_2(y) : \{x\} \times V \subseteq U\},$$

$$\tau(x, y) = \{U \subseteq X \times Y \mid \exists V \in \tau_1(x) \exists W \in \tau_2(y) : V \times W \subseteq U\}$$

# Full Product of Frames

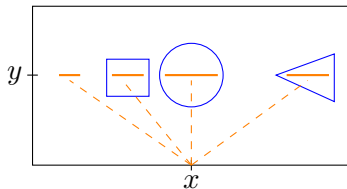
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# Multimodal Logics

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## Definition

Let  $L_1$  and  $L_2$  be two unimodal logics. Then the *fusion* ( $\otimes$ ) of these logics is

$$L_1 \otimes L_2 = K_2 + L_1^{\Box \rightarrow \Box_1} + L_2^{\Box \rightarrow \Box_2}$$

Furthermore the full product ( $\times_n^+$ ) is defined as

$$L_1 \times_n^+ L_2 = \text{Log}\{\mathcal{X} \times_n^+ \mathcal{Y} \mid \mathcal{X} \models L_1 \text{ and } \mathcal{Y} \models L_2\}$$

# Main Theorem

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## Theorem

*Let  $\Lambda$  be a unimodal logic with  $\Lambda \in \{T, D\}$ . Then the following holds:*

$$\Lambda \times_n^+ \Lambda = \Lambda \otimes \Lambda \otimes \Lambda + \Box p \rightarrow \Box_1 \Box_2 p \wedge \Box_2 \Box_1 p$$

In the following we will sketch the proof for  $\Lambda = T$ .

We denote  $T \otimes T \otimes T + \Box p \rightarrow \Box_1 \Box_2 p \wedge \Box_2 \Box_1 p$  as  $T_3m$

# Proof steps

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- Idea: create an infinite branching and depth Kripke tree with three relations (called  $T_{\omega,\omega,\omega}^{r,m}$ ) with  $\text{Log}(T_{\omega,\omega,\omega}^{r,m}) = T_3m$

$$T \otimes T \otimes T + (\Box p \rightarrow \Box_1 \Box_2 p \wedge \Box_2 \Box_1 p) = T_3m$$

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- construct a neighbourhood product  $N_{\omega}^r \times_n^+ N_{\omega}^r$  with  $\text{Log}(N_{\omega}^r) = T$  s.t. there is surjective bounded morphism from  $N_{\omega}^r \times_n^+ N_{\omega}^r$  to  $N(T_{\omega,\omega,\omega}^{r,m})$

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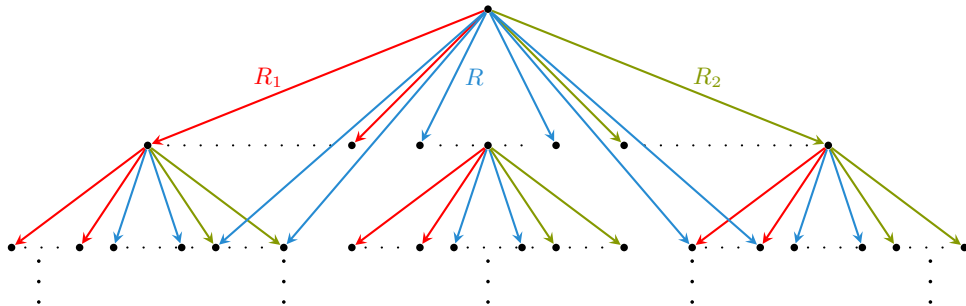
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- Keynote: For any Kripke frame  $F = (W, R_1, \dots, R_n)$ , we can construct a neighborhood frame  $N(F)$ , i.e. for any  $w \in W$ :  $\tau_i(w) = \{U \mid R_i(w) \subseteq U \subseteq W\}$  where  $\text{Log}(F) = \text{Log}(N(F))$

$$T \otimes T \otimes T + (\Box p \rightarrow \Box_1 \Box_2 p \wedge \Box_2 \Box_1 p) = T_3m$$

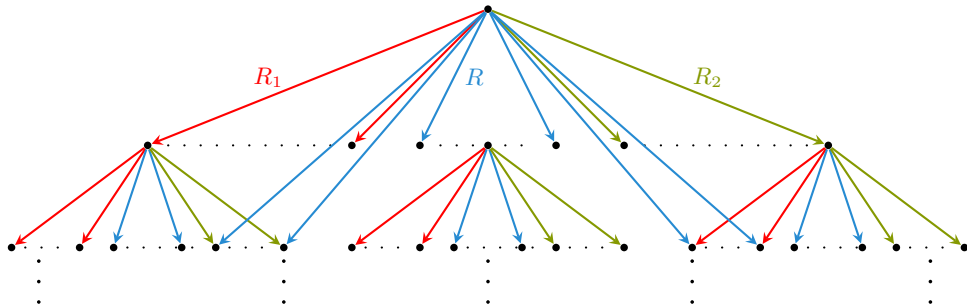
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- Note:  $\Box p \rightarrow \Box_1 p \wedge \Box_2$  is derivable
- $\text{Log}(T_{\omega,\omega,\omega}^{r,m}) \subseteq T_3m$  by standard techniques

# Definition of the neighborhood frame $N_w^r$

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## Definition

Let  $T_\omega^r = (\mathbb{N}^*, R)$  be the infinite branching depth tree. Then

$$X = \{a_0a_1a_2... \in (\mathbb{N} \cup \star) \mid \exists N \in \mathbb{N} \forall k \geq N : a_k = \star\}$$

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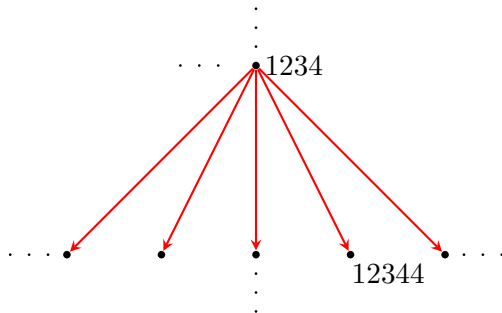
is the set of pseudo infinite sequences. Let  $\alpha = a_0a_1a_2... \in X$ . Then

$$U_k(\alpha) = \{\beta \in X \mid \text{forg}(\alpha)R\text{forg}(\beta) \text{ and } \alpha \upharpoonright_m = \beta \upharpoonright_m, \text{ with } m = \max(k, st(\alpha))\}$$

where  $\text{forg}(\alpha)$  deletes all  $\star$  and  $ls(\alpha) = \min\{N \mid \forall k \geq N : a_k = \star\}$

## Example of $U_k(\alpha)$

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Let  $\alpha = \star\star 12 \star 3 \star 4 \star\star^\omega \in X$ . Then  $ls(\alpha) = 9$ . Let's consider  $U_9(\alpha)$ . Then

$$\star\star 12 \star 3 \star 4 \star 4 \star\star^\omega \in U_9(\alpha)$$

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We define  $N_\omega^r = (X, \tau)$  where  $\tau : X \rightarrow 2^{2^X}$  with

$$\tau(\alpha) = \{U \subseteq X \mid \exists k \in \mathbb{N} : U_k(\alpha) \subseteq U\}$$

Note:  $\tau$  is a filter and there is no smallest neighborhood

# Surjective bounded morphism

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Let  $N_{\omega}^r \times_n^+ N_{\omega}^r = (X \times X, \tau_1, \tau_2, \tau)$  and  $N(T_{\omega, \omega, \omega}^{r, m}) = ((\mathbb{N} \times \{0, 1, 2\} \cup \{\star\})^*, \sigma_1, \sigma_2, \sigma)$



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## Theorem

*There is a surjective bounded morphism  $g : N_\omega^r \times_n^+ N_\omega^r \rightarrow N(T_{\omega, \omega, \omega}^{r, m})$ .*

Note:  $\text{Log}(N_w^r) = T$  and hence  $\text{Log}(N_w^r \times_n^+ N_w^r) \in T \times_n^+ T$

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|          |           |         |   |         |                |
|----------|-----------|---------|---|---------|----------------|
| $\alpha$ | $= \star$ | 1       | 2 | $\star$ | $\star^\omega$ |
| $\beta$  | $= 3$     | $\star$ | 7 | $\star$ | $\star^\omega$ |

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|                     |            |          |                |         |                |
|---------------------|------------|----------|----------------|---------|----------------|
| $\alpha$            | $= \star$  | $1$      | $2$            | $\star$ | $\star^\omega$ |
| $\beta$             | $= 3$      | $\star$  | $7$            | $\star$ | $\star^\omega$ |
| $g'(\alpha, \beta)$ | $= (3, 2)$ | $(1, 0)$ | $(h(7, 2), 0)$ | $\star$ | $\star^\omega$ |

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| $g'(\alpha, \beta)$ | $= (3, 2)$ | $(1, 0)$ | $(h(7, 2), 0)$ | $\star$ | $\star^\omega$ |

Then  $g(\alpha, \beta) = (3, 2)(1, 0)h((7, 2), 0)$

# Surjective bounded morphism

Let  $N_\omega^r \times_n^+ N_\omega^r = (X \times X, \tau_1, \tau_2, \tau)$  and  $N(T_{\omega, \omega, \omega}^{r, m}) = ((\mathbb{N} \times \{0, 1, 2\}^*, \sigma_1, \sigma_2, \sigma)$

## Theorem

*There is a surjective bounded morphism  $g : N_\omega^r \times_n^+ N_\omega^r \rightarrow N(T_{\omega, \omega, \omega}^{r, m})$ .*

- surjectivity and
- $R(g(\alpha, \beta)) \subseteq g(U_k(\alpha) \times U_j(\beta)) \subseteq g(U)$  for some  $k, j \in \mathbb{N}$

Note:  $\tau(x, y) = \{U \mid \exists k, j \in \mathbb{N} : U_k(x) \times U_j(y) \subseteq U\}$ ,  $\sigma(x) = \{U \mid R(x) \subseteq U\}$ .

2nd cond: for all  $1 \leq i \leq k$ , for all  $w \in W$  and all  $U \in \tau_i(w)$ , there is a  $V \in \sigma_i(f(w))$  with  $V = f(U)$

# Surjective bounded morphism

$$g(\alpha, \beta) = (3, 2)(1, 0)h((7, 2), 0) \textcolor{red}{R} (3, 2)(1, 0)h((7, 2), 0) (\textcolor{red}{h(6, 7), 0})$$

|                     |            |          |                |         |                                 |                |
|---------------------|------------|----------|----------------|---------|---------------------------------|----------------|
| $\alpha'$           | $= \star$  | 1        | 2              | $\star$ | $\textcolor{red}{6}$            | $\star^\omega$ |
| $\beta'$            | $= 3$      | $\star$  | 7              | $\star$ | $\textcolor{red}{7}$            | $\star^\omega$ |
| <hr/>               |            |          |                |         |                                 |                |
| $g'(\alpha, \beta)$ | $= (3, 2)$ | $(1, 0)$ | $(h(7, 2), 0)$ | $\star$ | $\textcolor{red}{(h(6, 7), 0)}$ | $\star^\omega$ |

Claim:  $R(g(\alpha, \beta)) \subseteq g(U_k(\alpha) \times U_j(\beta)) \subseteq g(U)$  for some  $k, j \in \mathbb{N}$



# Surjective bounded morphism

Let  $N_\omega^r \times_n^+ N_\omega^r = (X \times X, \tau_1, \tau_2, \tau)$  and  $N(T_{\omega, \omega, \omega}^{r, m}) = ((\mathbb{N} \times \{0, 1, 2\} \cup \{\star\})^*, \sigma_1, \sigma_2, \sigma)$

## Theorem

be two There is a surjective bounded morphism  $g : N_\omega^r \times_n^+ N_\omega^r \rightarrow N(T_{\omega, \omega, \omega}^{r, m})$ . Note:  $\text{Log}(N_w^r) = T$  and hence  $N_w^r \times_n^+ N_w^r \in T \times_n^+ T$

Pick  $m > \max\{st(\alpha), st(\beta)\}$  and set  $U := U_m(\alpha) \times U_m(\beta)$ . Then

$$g(U) = g(U_m(\alpha) \times U_m(\beta)) \subseteq R(g(\alpha, \beta)) \subseteq V$$

Note:  $\tau(x, y) = \{U \mid \exists k, j \in \mathbb{N} : U_k(x) \times U_j(y) \subseteq U\}$ ,  $\sigma(x) = \{U \mid R(x) \subseteq U\}$

3rd cond: for all  $1 \leq i \leq k$ , for all  $w \in W$  and all  $V \in \sigma_i(f(w))$ , there exists  $U \in \tau_i(w)$  such that  $f(U) \subseteq V$

# Future works

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